WiDFS 3.0 Reproduction Instructions

1. Folder Structure

(1) Delay_Doppler_Validation/

- Contains scripts for extracting Delay, Doppler, and Micro-Doppler features from single-target human motion datasets.
- These results correspond to *Section VIII.A (Delay–Doppler Feature Extraction Quality)* in the paper.

(2) Activity_Recognition_Validation/

- Contains scripts for validating the recognition performance of our WiDFS 3.0 scheme on the Widar 3.0 gesture recognition dataset.
- These results correspond to *Section VIII.B (Single-Target Sensing Performance*) in the paper.

2. Python Environment Setup

(1) Install Python 3.10 – 3.12

(2) Navigate to the project root directory and install dependencies:

pip install -r requirements.txt

3. Running the Programs

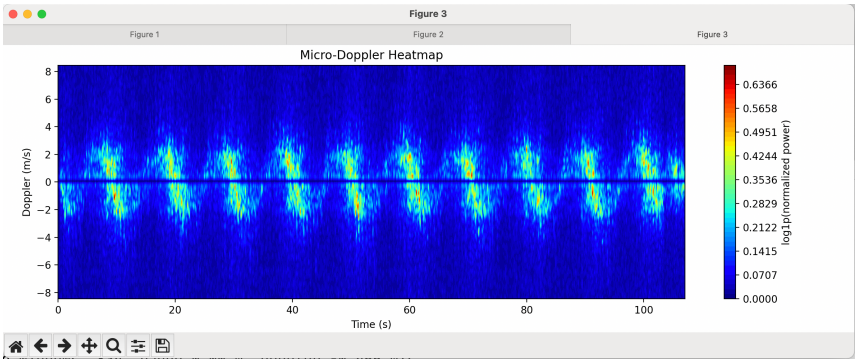
(1) Delay–Doppler Validation

Run: **python WiDFS3_Feature_Extractor.py**

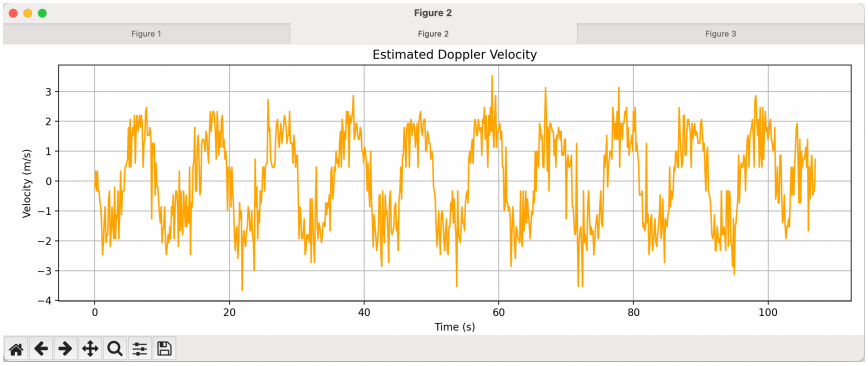
Output:

- Extracted Micro-Doppler spectrograms
- Delay and Doppler features

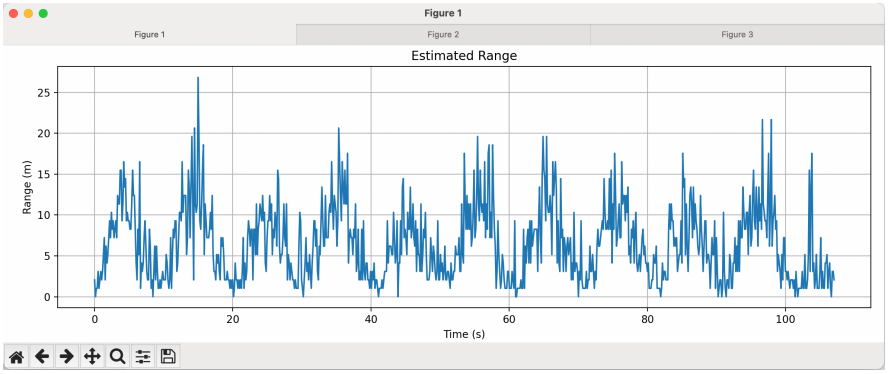
Micro-Doppler



Doppler Velocity

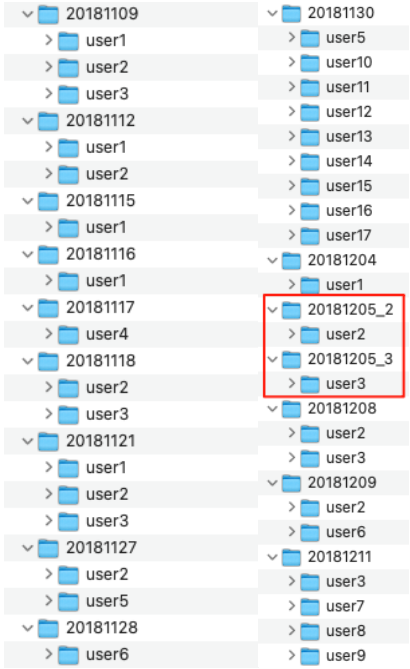


Range



4.2 Activity_Recognition_Validation

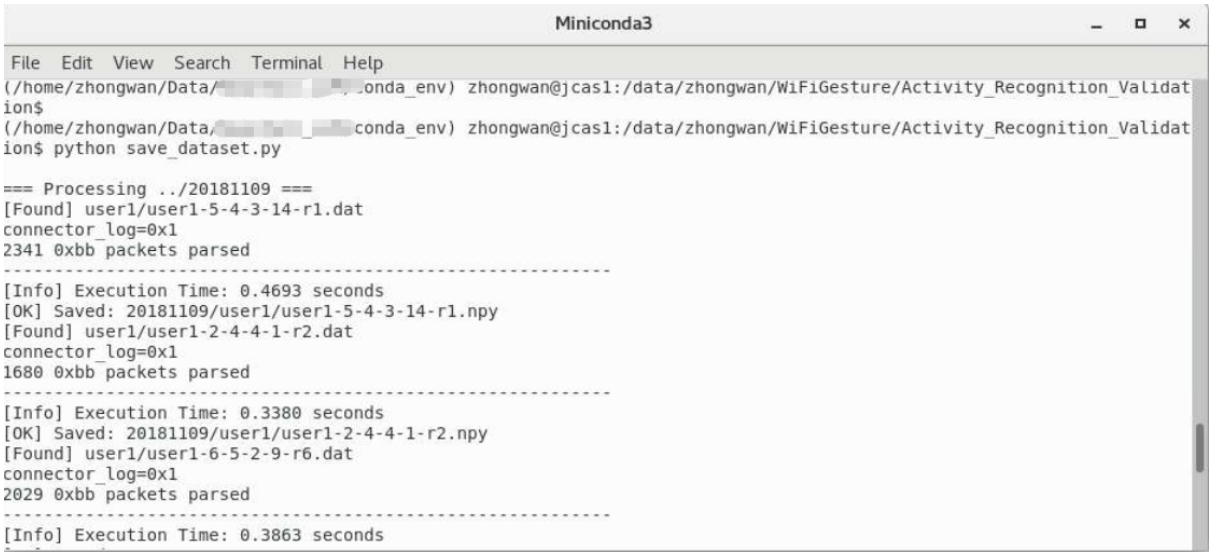
(1) Feature Generation (~42.3 GB size)



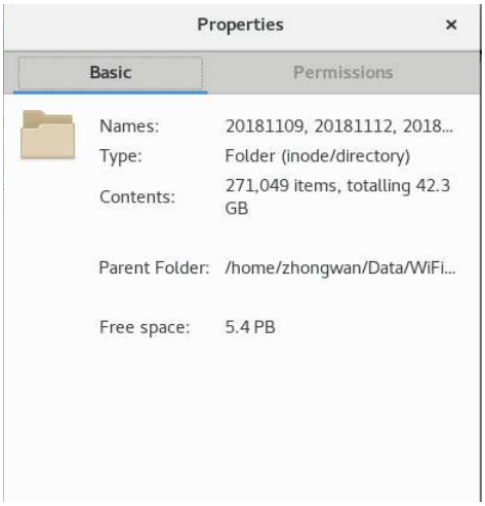
- Split the original folder 20181205 in Widar3.0 into two subfolders:
  - 20181205_2 containing only user2
  - 20181205_3 containing only user3
- In the same directory as the Widar3.0 dataset, place the Activity_Recognition_Validation folder. Then navigate into this folder and run:

python save_dataset.py

On our server, the feature extraction time for each .dat file (with 3 Rx antennas) is about 0.4 seconds.

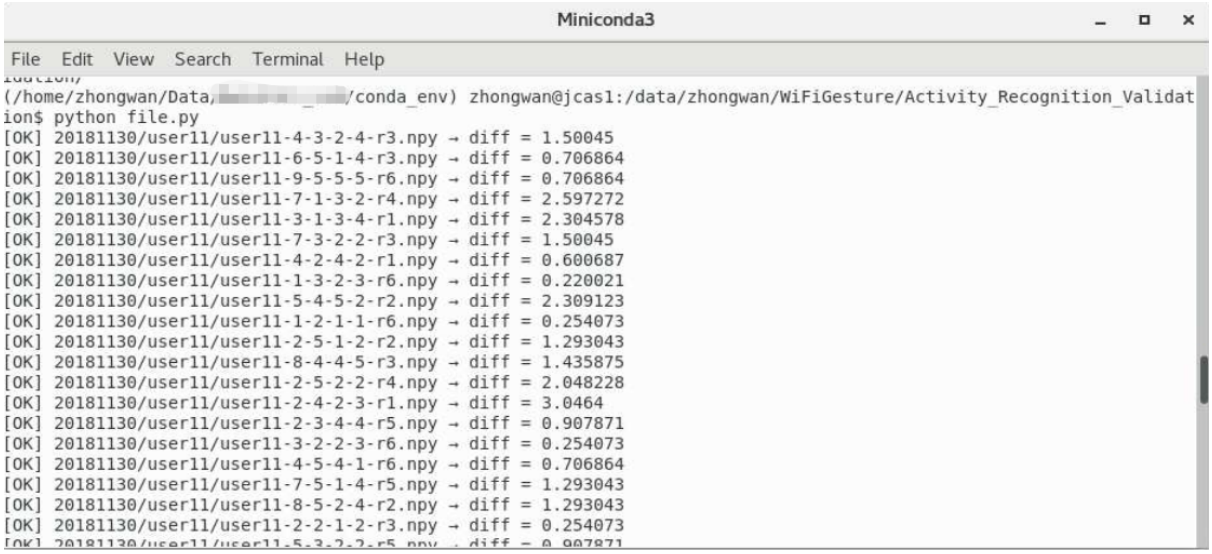


- After several hours, features will be generated. The feature directory structure will mirror that of the original Widar 3.0 CSI dataset.



(2) CSV Dataset Generation

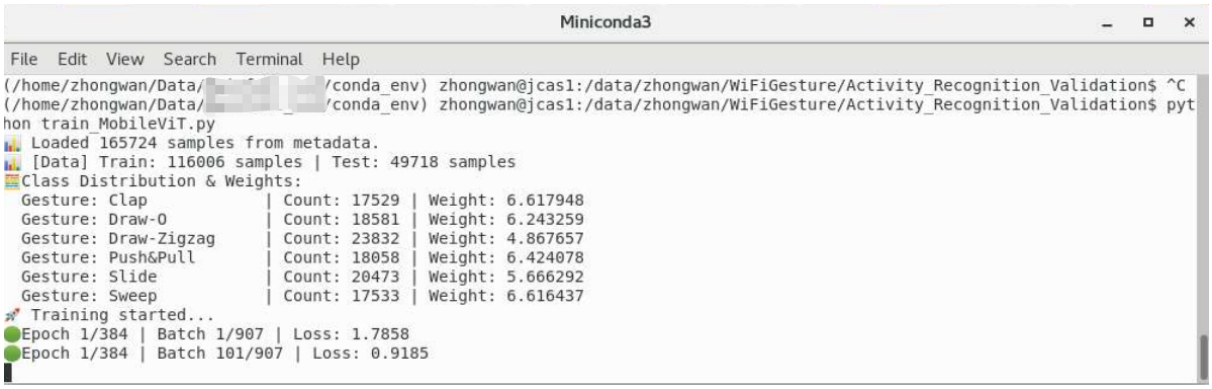
Run: **python file.py**



This will produce the dataset file gesture_metadata.csv.

(3) Model Training and Testing (70%–30% split)

Run: **python train_MobileVit.py**



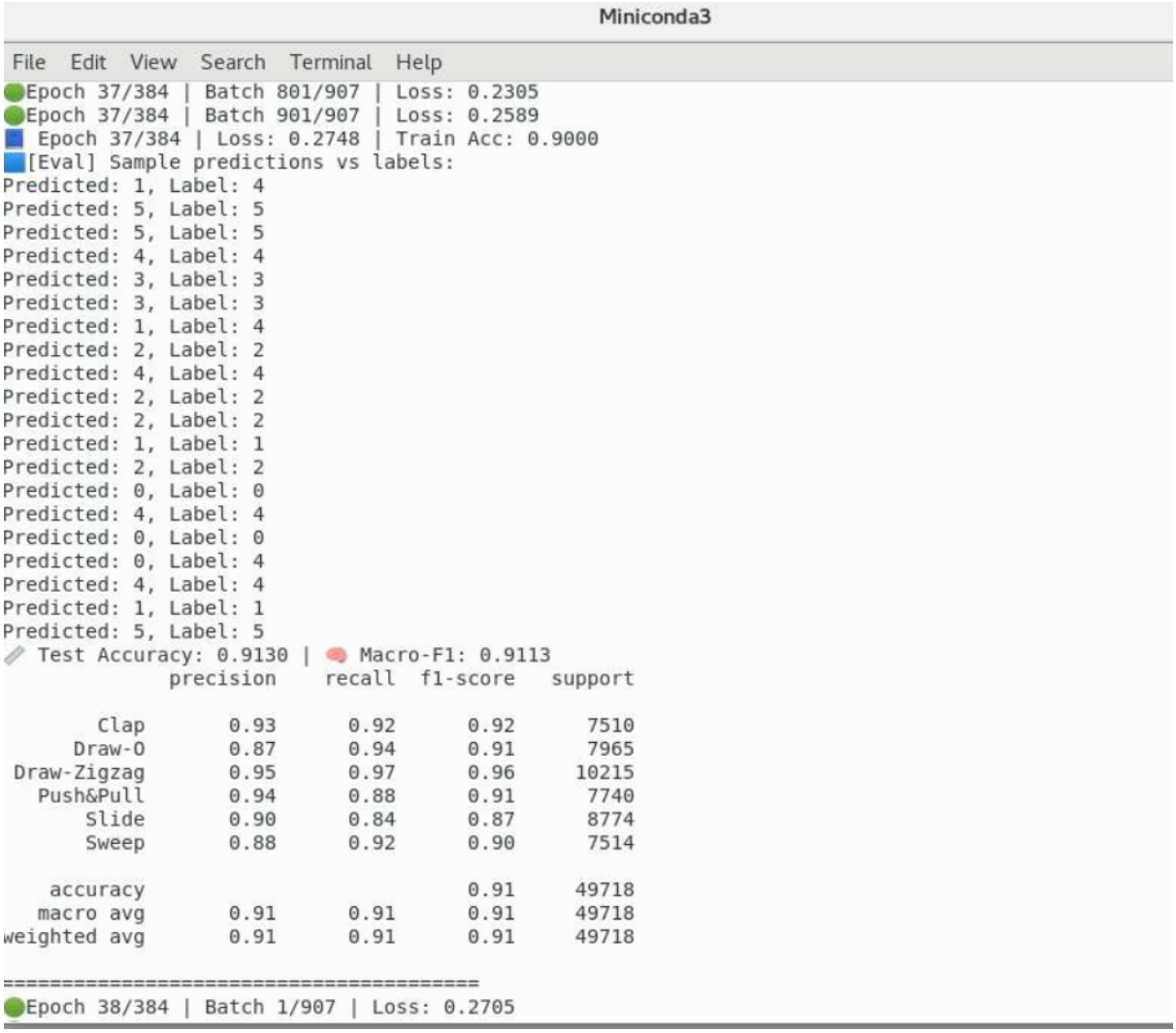
Miniconda3

File Edit View Search Terminal Help

```
🟢Epoch 20/384 | Batch 601/907 | Loss: 0.2517
🟢Epoch 20/384 | Batch 701/907 | Loss: 0.2996
🟢Epoch 20/384 | Batch 801/907 | Loss: 0.3697
🟢Epoch 20/384 | Batch 901/907 | Loss: 0.3885
🟦 Epoch 20/384 | Loss: 0.3272 | Train Acc: 0.8802
🟦[Eval] Sample predictions vs labels:
Predicted: 0, Label: 4
Predicted: 0, Label: 0
Predicted: 5, Label: 5
Predicted: 2, Label: 2
Predicted: 3, Label: 3
Predicted: 0, Label: 0
Predicted: 2, Label: 2
Predicted: 4, Label: 4
Predicted: 5, Label: 5
Predicted: 1, Label: 1
Predicted: 0, Label: 5
Predicted: 0, Label: 0
Predicted: 3, Label: 3
Predicted: 4, Label: 4
Predicted: 2, Label: 2
Predicted: 5, Label: 5
Predicted: 5, Label: 5
Predicted: 3, Label: 3
Predicted: 0, Label: 0
Predicted: 5, Label: 5
📝 Test Accuracy: 0.8983 | 🧠 Macro-F1: 0.8963
              precision    recall  f1-score   support

        Clap       0.90      0.93      0.91      7510
      Draw-O       0.83      0.95      0.88      7965
 Draw-Zigzag       0.96      0.95      0.96     10215
   Push&Pull       0.93      0.87      0.90      7740
       Slide       0.88      0.81      0.85      8774
       Sweep       0.88      0.88      0.88      7514

    accuracy                           0.90     49718
   macro avg       0.90      0.90      0.90     49718
weighted avg       0.90      0.90      0.90     49718
```

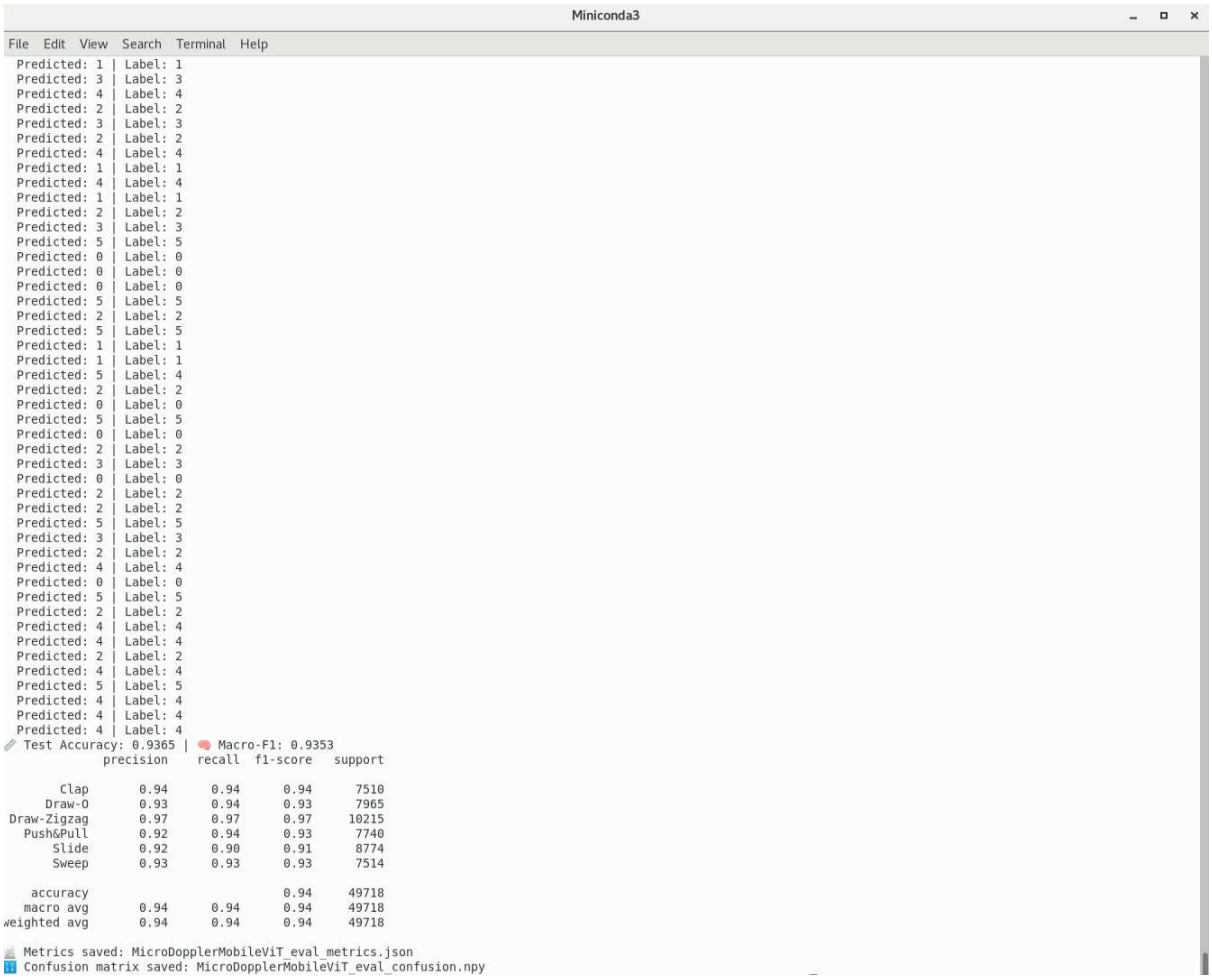


The training and testing process will start automatically. A new folder saved_models_gesture will be created, containing trained models and result logs.

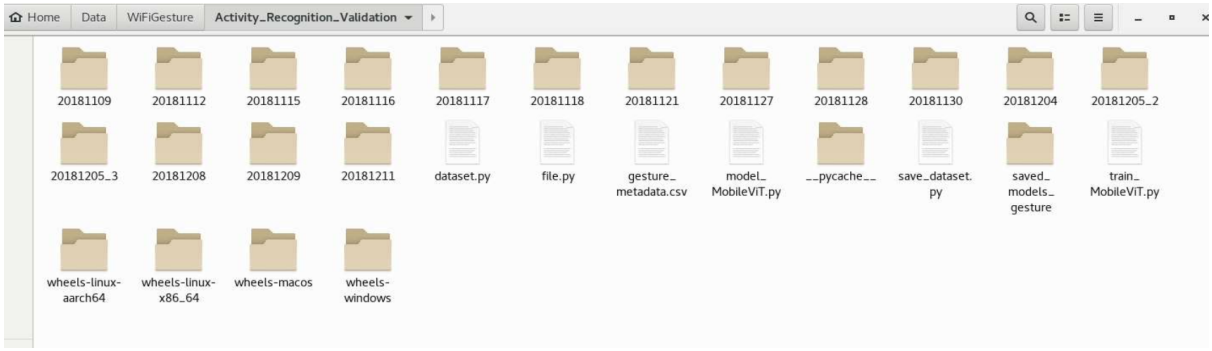
(4) Model Testing Only (70%–30% split)

Using the trained model, you can directly evaluate performance.

Run: **python evaluate_only.py**



In addition, the main folder contains both the extracted features and the source code.



The submitted folder saved_models_gesture contains our trained MobileViT models, along with the corresponding confusion matrices and the metrics.json file.

5. Contact

If you encounter any problems during reproduction, please contact:

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